[redacted]

Li-ion Batteries For Solar Setups

In-built lithium-ion batteries for solar streetlights in Assam have reduced system losses by minimising wiring

Zonstreet with in-built lithium battery in Assam before flooding (left) and post flooding (right) (Courtesy: Dr Uppal)

Batteries play a crucial role in solar setups. There are different types of battery technologies available for solar energy storage. Operational advantages and costs associated with each of these options directly impact the payback period of solar setups. While lead-acid batteries are undergoing substantial improvements with refinement of technology, lithium-ion batteries keep going strong in the market.

Conventional flooded lead-acid batteries are inexpensive. Dr Kushant Uppal, founder and CEO, Intelizon Energy, shares, "Lead-acid batteries are heavy and large in size. This creates some challenges in commercial applications like solar streetlights, where these batteries are installed externally on the pole. As lead-acid batteries need regular maintenance, these need to be installed at accessible heights. This makes them vulnerable to theft. Maintenance also adds to operational expenses."

Lead-acid batteries can discharge up to 80 per cent. That means these cannot be drained out completely when in use. Moreover, the time taken for charging is quite high at around 9-11 hours.

However, these batteries have some positive aspects too. Ankit Bajpai, executive director, EPC, Trisun Sant India, says, "The cost of maintaining lead-acid batteries is nominal. If maintained well, these batteries can provide a longer life of operation, which can go up to five years per battery."

The new normal in the market are lithium-ion and lithium ferrophosphate batteries. Uppal explains, "Lithium-ion batteries have a high energy density and are maintenance-free. Therefore there are no running costs associated with these. Moreover, they weigh just one-third of what an equivalent capacity of lead-acid battery will weigh. This makes them easy to install and therefore less prone to theft and damage."

Ankit says, "Lithium batteries allow full charge consumption during usage. Moreover, these batteries have a lifespan of 3 to 3.5 years, which is higher than majority of lead-acid batteries. However, lithium-ion batteries cost at least 25 per cent higher than lead-acid batteries."

Among other battery technologies, gel batteries are making their presence felt all around. These are valve-regulated lead-acid (VLRA) batteries, which offer significant advantages over their conventional lead-acid counterparts.

Biju Bruno, managing director, Greenvision Technologies, says, "Gel batteries are part of the lead-acid ecosystem, but are well-developed and mature. While conventional flooded tubular lead-acid batteries cost 30 per cent less, gel batteries provide better operations with less maintenance."

However, gel batteries have low thermal tolerance, which means these can malfunction if overheated. Additionally, they are highly sensitive to overcharging and may result in operational failures.

Lithium-ion batteries cost much higher than conventional lead-acid or even gel batteries. Bruno says, "Typically, lithium-ion battery packs cost approximately ₹ 25,000 per kW compared to ₹ 10,000 for gel batteries."

However, given their advantages, lithium-ion batteries are still the best bet.

Case Study: Li-ion batteries for solar LED streetlights in Assam

Assam receives heavy rainfall throughout the year. Choice of the battery proved to be a challenge for solar LED streetlights installed in a remote region of the state. The adverse weather conditions shortened the expected lifespan of lead-acid batteries used, requiring their replacement every six months. This greatly added to the running expenses.

There was yet another big challenge. Uppal says, "On cloudy days, solar energy generation drops to below 50 per cent. Along with that, there was major energy loss due to the long cables connecting the battery to the panel and use of low-efficiency (typically 70-75 per cent) charge-controllers. Also, it was critical to have a separate solar panel, which is installed facing south for the best charging efficiency."

To address these challenges, Intelizon Energy provided streetlights with in-built lithium-ion batteries. They call this solution Zonstreet.

Uppal explains, "The lithium-ion setup reduced system losses by minimising wiring. We implemented high-efficiency (>95 per cent) charge-controllers, which further improved the figure. Lithium technology enabled the right battery sizing, which